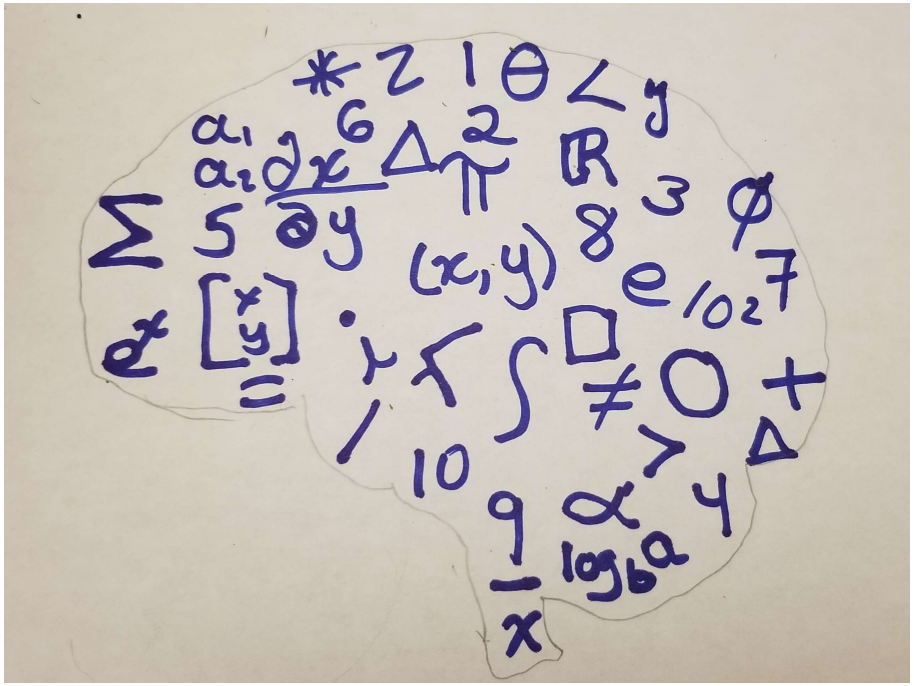
Symbols, conventions, and invisible numbers

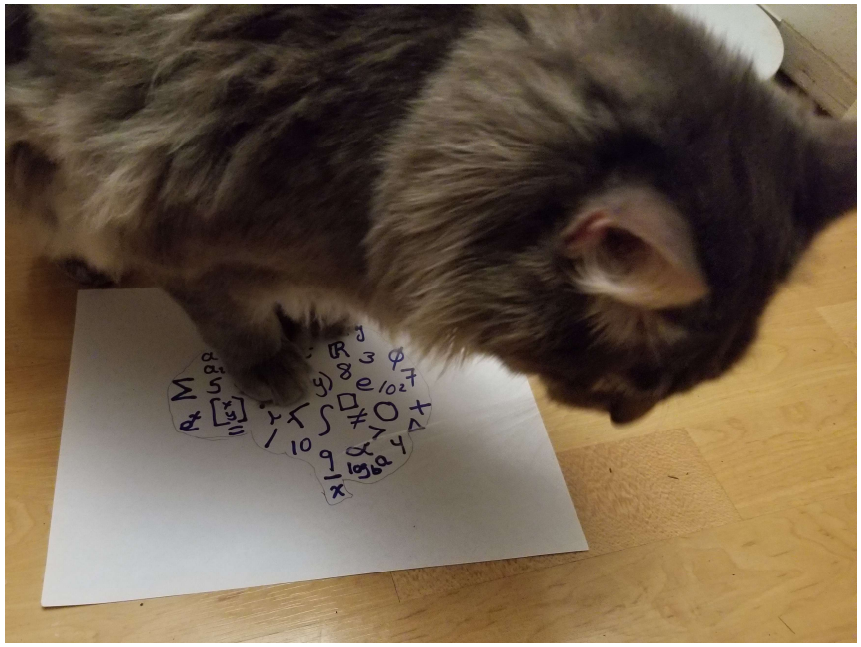
There are lots of symbols used to write and communicate mathematical concepts, ideas, proofs, and problems.

Some people have difficulty with symbolic reasoning and interpreting symbols.

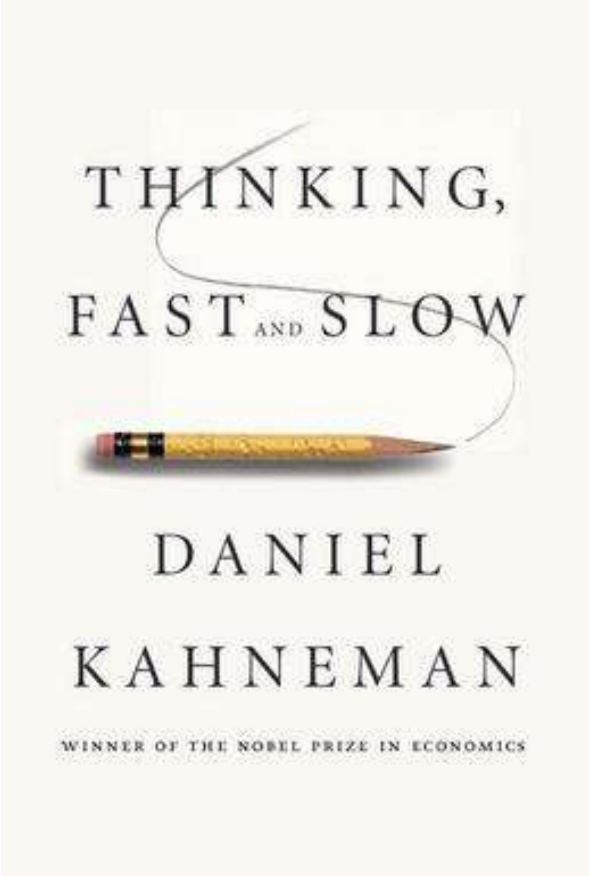


You can train your brain!

If you have this challenge, speak up so that we can figure out how to train your brain for quick recognition of the symbols. Doc wants to help.



Thinking, Fast and Slow by Daniel Kahneman



Daniel Kahneman is a prize-winning behavioral economist who talks about the fast system 1 and the slower system 2 thinking pathways.

I use his observations and theories in developing this curriculum.

https://youtu.be/amnnX6grUls is a video that explains the book.

Don’t forget to stay in the Just Right Challenge zone!

The Occupational Therapist Jean Ayers came up with the concept of the *Just Right Challenge*.

When things are too easy, they are boring.

When things are too hard, they are overwhelming.

The Just Right Challenge is in that sweet spot of being interesting and exciting, while still creating challenge to learn and grow.

When you are moving through new concepts and doing practice, don’t forget to stay in the Just Right Challenge zone!

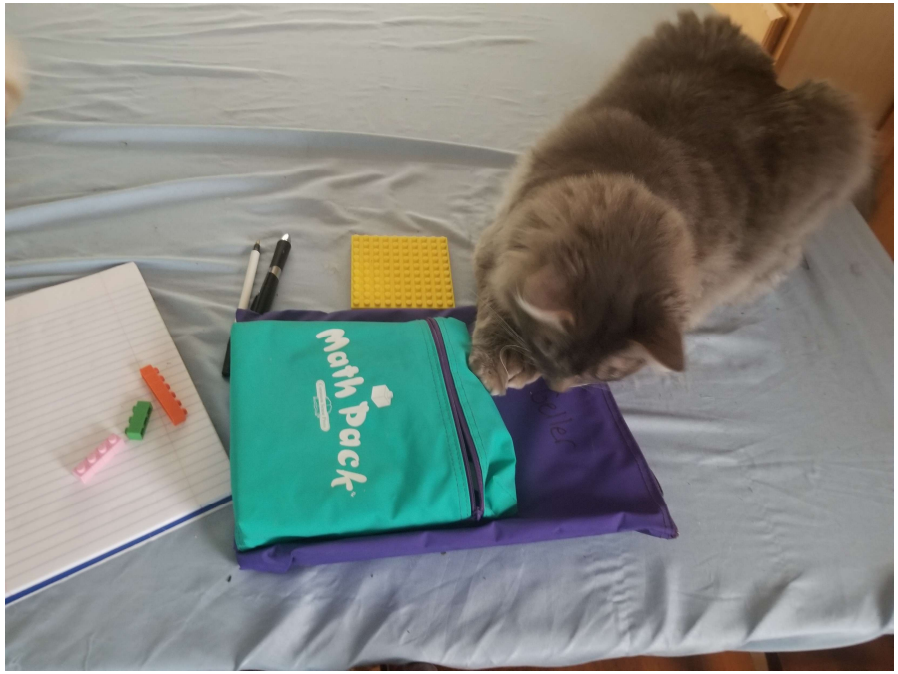
When you start, it is all the slow system 2 and then practice can get you into system 1 for some problems.

Knowing when problems need to go into system 2, slow, deliberative thinking is very helpful.

Knowing when you need to use tools and having quick access to your tools is also very helpful.

Doing math problems in your head while you are driving is not a good idea!

Supposedly, going over concepts before bed can help get them into memory.

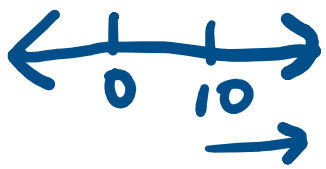


If right and left are a problem, start with up and down

You may have noticed that I include up and down representations of a single number line. I get right and left confused so find it easier to use up and down for a single number line.

Remember that the English decimal system gets bigger or has greater place values to the left, but the horizontal number line gets bigger to the right!

432.1



Conventions

There are many conventions that are used so that everyone can communicate more clearly.

The order of operations is a great example of a convention that people follow. $2(x+1)+3$

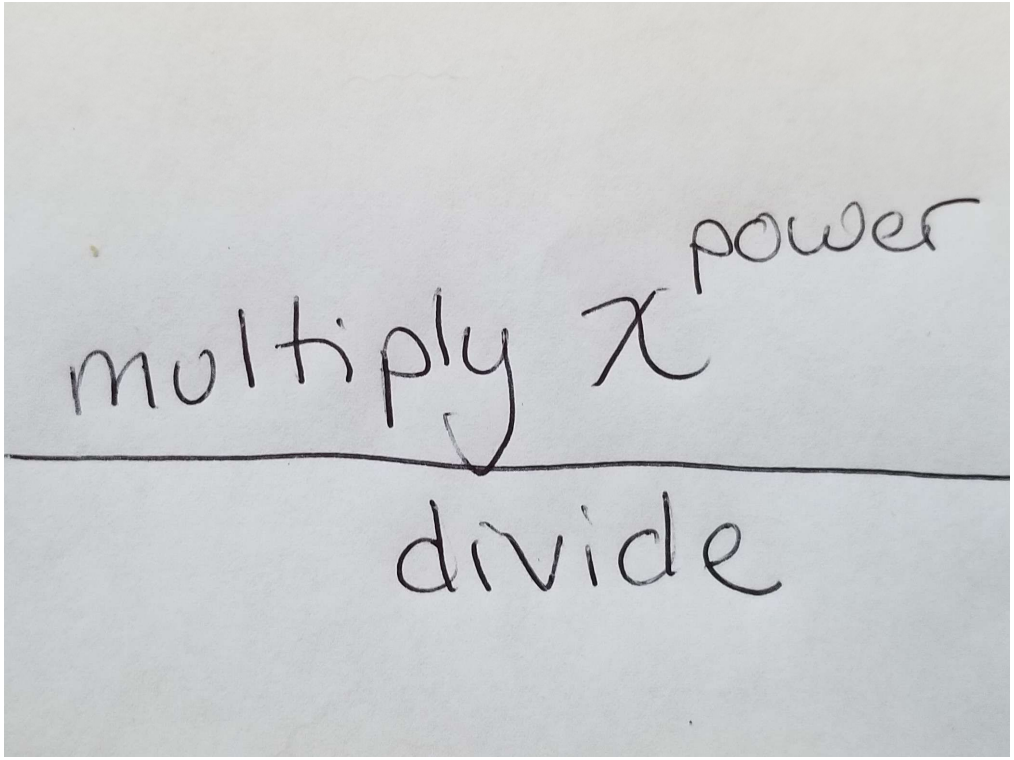
Parentheses

Exponents

Multiplication/Division

Addition/Subtraction

Placement of numbers and variables has different meanings.



For programming

+ plus for addition

- minus for subtraction

* multiplication sign it is implied 2x is 2 times x, but most programming languages need 2*x

/ division bar is used for division. Need to use parentheses with the bar

6/(2+1)

$\frac{6}{2+1}$ When you write division by hand, the bar is a grouping symbol

$\frac{6}{(x+1)}$

** or ^ are the symbols for exponentiation depending on the language

3**2 or 3^2 is 3^2 You have to learn the symbol for the particular language.

Exponentiation is multiple multiplications

I will mention it and then we will come back to it much later.

The number you are multiplying is the base and how many times you multiply it is the exponent or power.

The exponent is written as a superscript.

$$base^{exponent}$$

Many computer languages use the **^** symbol but python uses ** for exponentiation. Since it is easy to access on a keyboard and makes sense, I will use that when I can.

Exponents have bases to a power.

2, 10, and *e* are the most common numbers used as bases.

Our base 10 number system has different names for the powers of 10:

10**1 or 10^1 10^1 is ten

10**2 or 10^2 is a hundred

10**3 or 10^3 is a thousand

10**6 or 10^6 is a million etc.

Binary is powers of 2

Polynomial functions or equations

Polynomial functions have the variable in the base.

This means that you are multiplying the variable by itself as many times as in the exponent.

x**1 or x^1 is x

x**2 or x^2 is x times x or xx or x*x

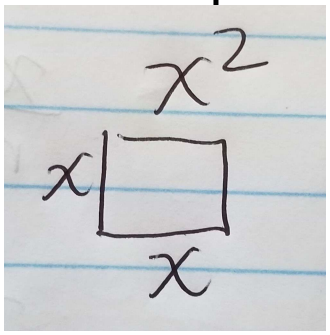
x**3 or x^3 is x times x times x or xxx or x*x*x

You can visualize the power or exponent as the number of dimensions.

There are special words for exponents of 2 and 3

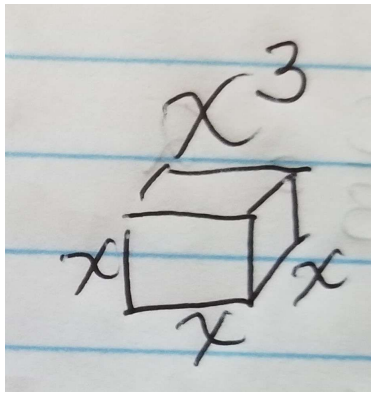
For a base to the second power, we say the base is squared.

x^2 is x squared which is x times x

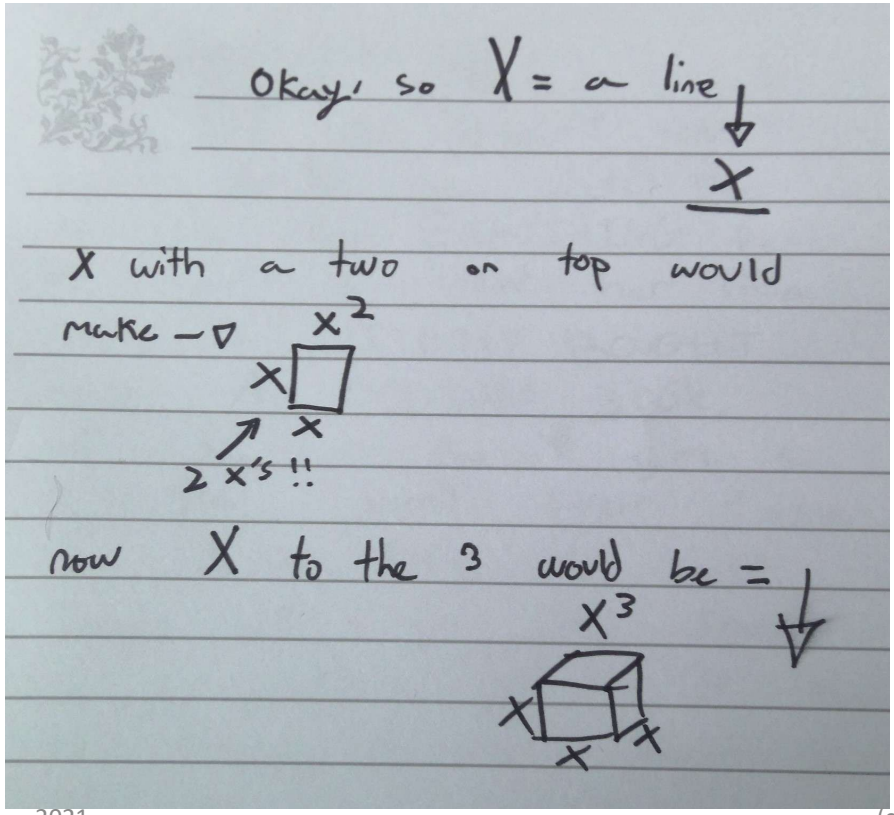


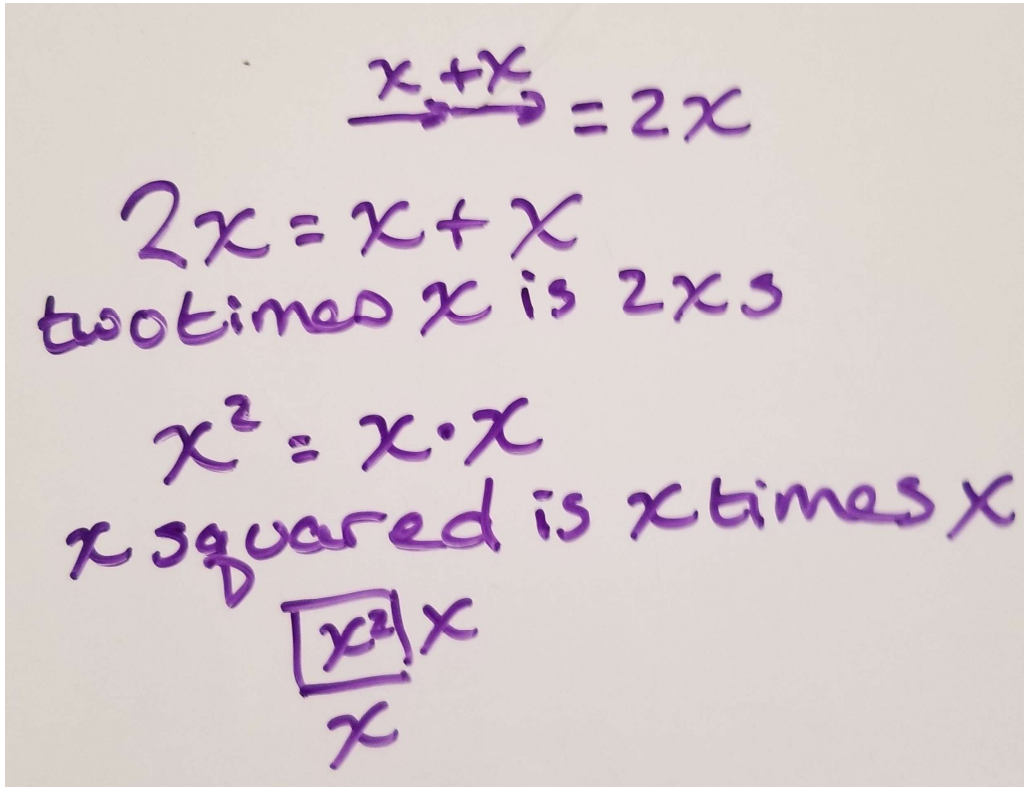
For a base to the third power, we say the base is cubed.

x^3 is x cubed which is x times x times x



Polynomial (variable to a counting number power) as dimensions Drawing ©ZMR





Functions have special names

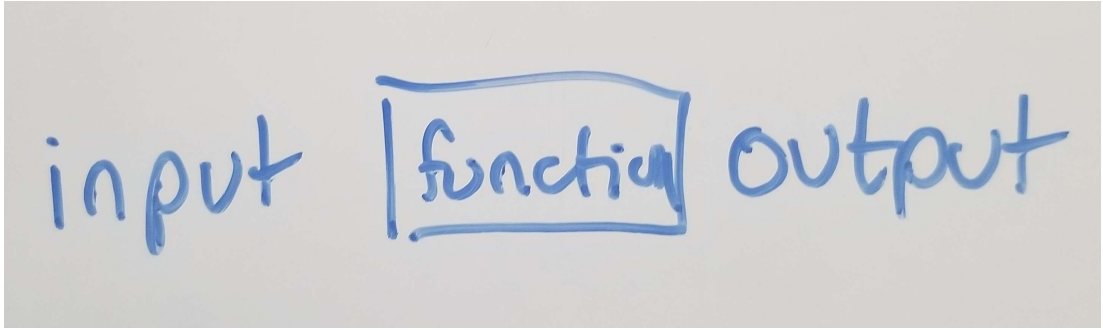
x^1 is a linear function x

x^2 is a quadratic function

From the Latin for four

x^3 is a cubic function

From the Greek



We will come back to that when we look at higher order functions. For now, we are still looking at one variable to the first power, or linear functions.

Exponential functions or equations have the variable in the exponent

10, and *e* most common bases for exponential functions.

They are written as 10^x and e^x and we will talk about them much later.

Notice that the variable is in the exponent and the base is a constant number.

We will come back to this much later!

$2^x \neq x^2$

x	x^2		2^x
0	$0^2 = 0 \cdot 0 = 0$		$2^0 = \frac{2}{2} = 1$
1	$1^2 = 1 \cdot 1 = 1$		$2^1 = 2 = 2$
2	$2^2 = 2 \cdot 2 = 4$	$=$	$2^2 = 2 \cdot 2 = 4$
3	$3^2 = 3 \cdot 3 = 9$		$2^3 = 2 \cdot 2 \cdot 2 = 8$
4	$4^2 = 4 \cdot 4 = 16$	$=$	$2^4 = 2 \cdot 2 \cdot 2 \cdot 2 = 16$
5	$5^2 = 5 \cdot 5 = 25$		$2^5 = 2 \cdot 2 \cdot 2 \cdot 2 \cdot 2 = 32$

x	x^3	3^x
1	$1^3 = 1$	$3^1 = 3$
2	$2^3 = 8$	$3^2 = 3 \cdot 3 = 9$
3	$3^3 = 27$	$3^3 = 3 \cdot 3 \cdot 3 = 27$

Subscripts

x+N=A

x-N=A

Unless N is zero, the Ns and the As need to be different in these two equations. We can use subscripts to represent numbers in a sequence or the same type of variable.

If c is the number of cats in each home, c_1 could be the number in one home and c_2 could be the number in a second home.

If we want x to be the same, we could write:

x+n_1= a_1

x- n_2= a_2 to indicate that they are different Ns and As

Placement matters

Different locations have different meanings.

The 1 refers to this being the first x in a series with multiple values of x. The 2 means that we square the x (multiply x*x). The 3 means that multiply by 3. The 4 means that we divide by 4. Notice the different locations.

The 1 is called a subscript and the 3 is a superscript in Word or PowerPoint.

$$\frac{3x_1^2}{4}$$

Invisible numbers and signs are implied.

When we write out math, there are lots of what I call invisible numbers and signs.

x is also +x with an invisible plus sign. We always put a minus sign for a negative number, but the plus sign is implied if it isn't there.

2x and 2(x+1) both have an invisible multiplication sign. If you type it into a computer or calculator, you will need to add the * multiplication sign.

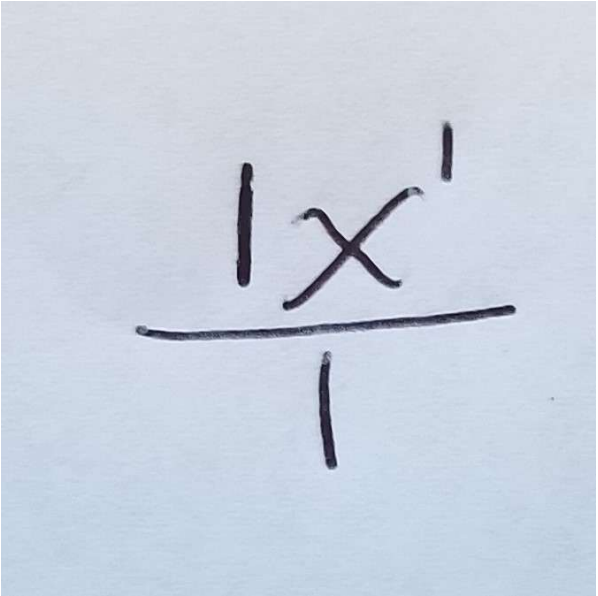
If you are using a horizontal bar for a fractional or rational expression, there are invisible parentheses, since the bar acts as a grouping symbol.

$$\frac{x+1}{2} = 3$$

$$\frac{x+1}{x+2} = \frac{(x+1)}{(x+2)}$$

Invisible numbers

when we write x, it is also:



Invisible terms

Later, we will see that we can also have invisible terms:

$$\begin{array}{r} 3x+0 \\ \underline{x+2} \\ 4x+2 \end{array}$$

$$\begin{array}{r} x^2 + 0\mathrm{x} + 3 \\ \underline{x^2 + 5\mathrm{x} + 1} \\ 2\,x^2 + 5\mathrm{x} + 4 \end{array}$$

Modulo operation

The % symbol is used to mean percent. x% is x/100

% is also used in many programming languages for the modulo operation.

Modular arithmetic, or clock arithmetic, is when you go around a certain number of times and then have a leftover. Like our twelve-hour clock system. 13:00 hours is the same as 1 pm on a twelve-hour clock. 13%12 is 1

This happens with twelve-hour clocks and later, when we do imaginary numbers. I wanted to point it out when we are talking about symbols. I think of this as the leftover operation.

In python x//y is the integer part of the division and x%y is the remainder.

`x % y`	remainder of `x / y`
`divmod(x, y)`	the pair `(x // y, x % y)`
`pow(x, y)`	*x* to the power *y*
`x ** y`	*x* to the power *y*